Release notes

Astra Control Service

NetApp
March 27, 2024

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Release notes

What's new with Astra Control Service

NetApp periodically updates Astra Control Service to bring you new features, enhancements, and bug fixes.

14 March 2024

New features and support

This is a minor bug fix release.

7 November 2023

New features and support

- **Backup and restore capabilities for applications with ontap-nas-economy driver-backed storage backends**: Enable backup and restore operations for `ontap-nas-economy` with some simple steps.
- **Astra Control Service support for on-premises Red Hat OpenShift Container Platform clusters**

  Add a cluster

- **Immutable Backups**: Astra Control now supports unalterable, read-only backups as an additional security layer against malware and other threats.
- **Introducing Astra Control Provisioner**

  With the 23.10 release, Astra Control introduces a new software component called Astra Control Provisioner that will be available to all licensed Astra Control users. Astra Control Provisioner provides access to a superset of advanced management and storage provisioning features beyond those that Astra Trident provides. These features are available to all Astra Control customers at no additional cost.

  - **Get started with Astra Control Provisioner**
    You can enable Astra Control Provisioner if you have installed and configured your environment to use Astra Trident 23.10.
  - **Astra Control Provisioner functionality**

    The following features are available with the Astra Control Provisioner 23.10 release:

    - **Enhanced storage backend security with Kerberos 5 encryption**: You can improve storage security by enabling encryption for the traffic between your managed cluster and the storage backend. Astra Control Provisioner supports Kerberos 5 encryption over NFSv4.1 connections from Red Hat OpenShift clusters to Azure NetApp Files and on-premise ONTAP volumes.
    - **Recover data using a snapshot**: Astra Control Provisioner provides rapid, in-place volume restoration from a snapshot using the `TridentActionSnapshotRestore` (TASR) CR.
    - **Backup and restore capabilities for applications with `ontap-nas-economy` driver-backed storage backends**: As described above.
- **Astra Control Service support for Red Hat OpenShift Service on AWS (ROSA) clusters**

  Add a cluster

- **Support for managing applications that use NVMe/TCP storage**
  Astra Control can now manage applications backed by persistent volumes that are connected using NVMe/TCP.
- **Execution hooks turned off by default**: Beginning with this release, execution hooks functionality can be enabled or disabled for additional security (it is disabled by default). If you have not yet created execution hooks for use with Astra Control, you need to enable the execution hooks feature to begin creating hooks. If you created execution hooks prior to this release, the execution hooks functionality stays enabled and you can use hooks as you would normally.

2 October 2023

New features and support

This is a minor bug fix release.

27 July 2023

New features and support

- Clone operations now support live clones only (current state of managed application). To clone from a snapshot or backup, use the restore workflow.

  Restore apps

26 June 2023

New features and support

- Azure Marketplace subscriptions are now billed per hour instead of per minute

  Set up billing

30 May 2023

New features and support

- Support for private Amazon EKS clusters

  Manage private clusters from Astra Control Service

- Support for selecting the destination storage class during restore or clone operations

  Restore apps

15 May 2023

New features and support

This is a minor bug fix release.

25 April 2023

New features and support

- Support for private Red Hat OpenShift clusters

Manage private clusters from Astra Control Service

- Support for including or excluding application resources during restore operations

  Restore apps

- Support for managing data-only applications

  Start managing apps

17 January 2023

New features and support

- Enhanced execution hooks functionality with additional filtering options

  Manage app execution hooks

- Support for NetApp Cloud Volumes ONTAP as a storage backend

  Learn about Astra Control

22 November 2022

New features and support

- Support for applications that span across multiple namespaces

  Define apps

- Support for including cluster resources in an application definition

  Define apps

- Enhanced progress reporting for your backup, restore, and clone operations

  Monitor running tasks

- Support for managing clusters that already have a compatible version of Astra Trident installed

  Start managing Kubernetes clusters from Astra Control Service

- Support for managing multiple cloud provider subscriptions in a single Astra Control Service account

  Manage cloud instances

- Support for adding self-managed Kubernetes clusters that are hosted in public cloud environments to Astra Control Service

  Start managing Kubernetes clusters from Astra Control Service

- Billing for Astra Control Service is now metered per namespace instead of per application

  Set up billing

- Support for subscribing to Astra Control Service term-based offers through the AWS Marketplace

Set up billing

Known issues and limitations

- Known issues for this release
- Known limitations for this release

7 September 2022

This release includes stability and resiliency enhancements for the Astra Control Service infrastructure.

10 August 2022

This release includes the following new features and enhancements.

- Improved application management workflow
  Improved application management workflows provide increased flexibility when defining applications managed by Astra Control.

  Manage apps

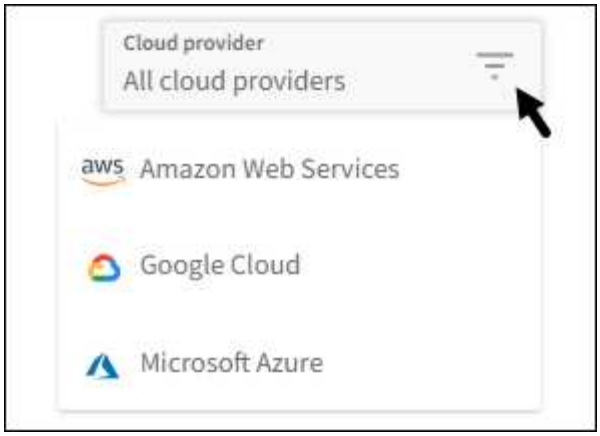
- Enhanced execution hooks
  In addition to pre- and post-snapshot execution hooks, you can now configure the following types of execution hooks:
  - Pre-backup
  - Post-backup
  - Post-restore

  Among other improvements, Astra Control now supports using the same script for multiple execution hooks.

  > The NetApp-provided default pre- and post-snapshot execution hooks for specific applications have been removed in this release. If you do not provide your own execution hooks for snapshots, Astra Control Service will take crash-consistent snapshots only beginning August 4, 2022. Visit the NetApp Verda GitHub repository for sample execution hook scripts that you can modify to fit your environment.

  Manage app execution hooks

- Cloud provider selection
  While reading the Astra Control Service documentation, you can now select your cloud provider at the top right of the page. You will see documentation relevant only to the cloud provider you select.



26 April 2022

This release includes the following new features and enhancements.

- Namespace role-based access control (RBAC)
  Astra Control Service now supports assigning namespace constraints to Member or Viewer users.

  Namespace role-based access control (RBAC)

- Bucket removal from Astra Control
  You can now remove a bucket from Astra Control Service.

  Remove a bucket

14 December 2021

This release includes the following new features and enhancements.

- New storage backend options

- In-place app restore
  You can now restore a snapshot, clone, or backup of an app in place by restoring to the same cluster and namespace.

  Restore apps

- Script events with execution hooks
  Astra Control supports custom scripts that you can run before or after you take a snapshot of an application. This enables you to perform tasks like suspending database transactions so that the snapshot of your database app is consistent.

  Manage app execution hooks

- Operator-deployed apps
  Astra Control supports some apps when they are deployed with operators.

  Start managing apps

5 August 2021

This release includes the following new features and enhancements.

- Astra Control Center
  Astra Control is now available in a new deployment model. *Astra Control Center* is self-managed software that you install and operate in your data center so that you can manage Kubernetes application lifecycle management for on-premise Kubernetes clusters.

  To learn more, go to the Astra Control Center documentation.

- Bring your own bucket
  You can now manage the buckets that Astra uses for backups and clones by adding additional buckets and by changing the default bucket for the Kubernetes clusters in your cloud provider.

  Manage buckets

2 June 2021

This release includes bug fixes and the following enhancements to Google Cloud support.

- Support for shared VPCs
  You can now manage GKE clusters in GCP projects with a shared VPC network configuration.
- Persistent volume size for the CVS service type
  Astra Control Service now creates persistent volumes with a minimum size of 300 GiB when using the CVS service type.

  Learn how Astra Control Service uses Cloud Volumes Service for Google Cloud as the storage backend for persistent volumes.

- Support for Container-Optimized OS
  Container-Optimized OS is now supported with GKE worker nodes. This is in addition to support for Ubuntu.

  Learn more about GKE cluster requirements.

15 April 2021

This release includes the following new features and enhancements.

- REST API
  The Astra Control REST API is now available for use. The API is based on modern technologies and current best practices.

  Learn how to automate application data lifecycle management using the REST API.

- Annual subscription
  Astra Control Service now offers a *Premium Subscription*.

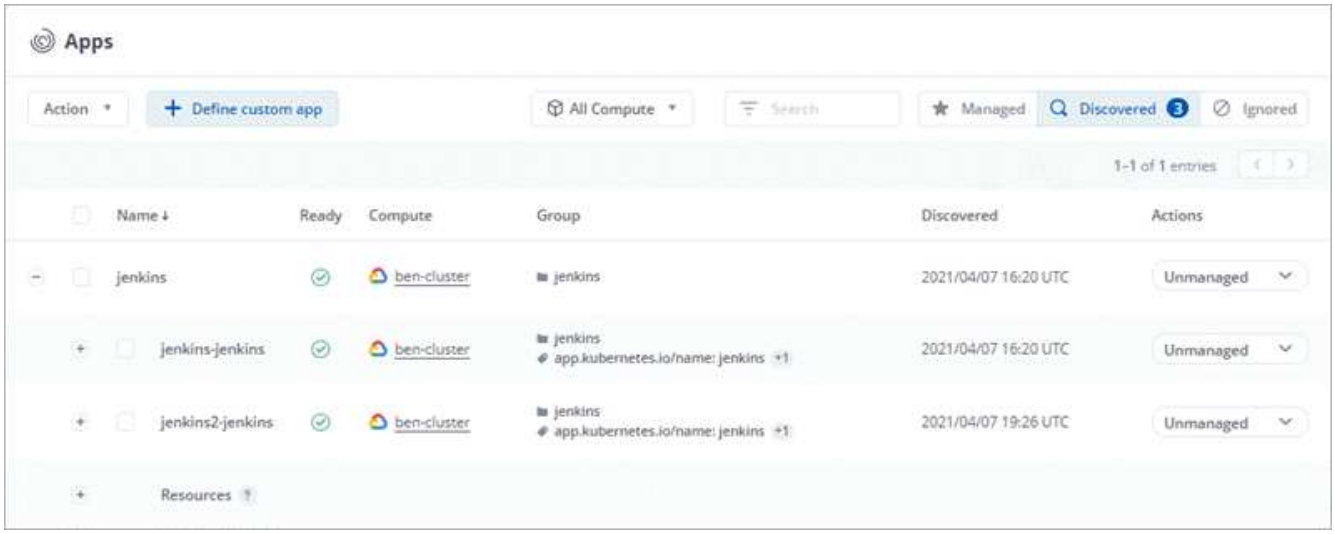
  Pre-pay at a discounted rate with an annual subscription that enables you to manage up to 10 apps per *application pack*. Contact NetApp Sales to purchase as many packs as needed for your organization—for example, purchase 3 packs to manage 30 apps from Astra Control Service.

If you manage more apps than allowed by your annual subscription, then you'll be charged at the overage rate of $0.005 per minute, per application (the same as Premium PayGo).

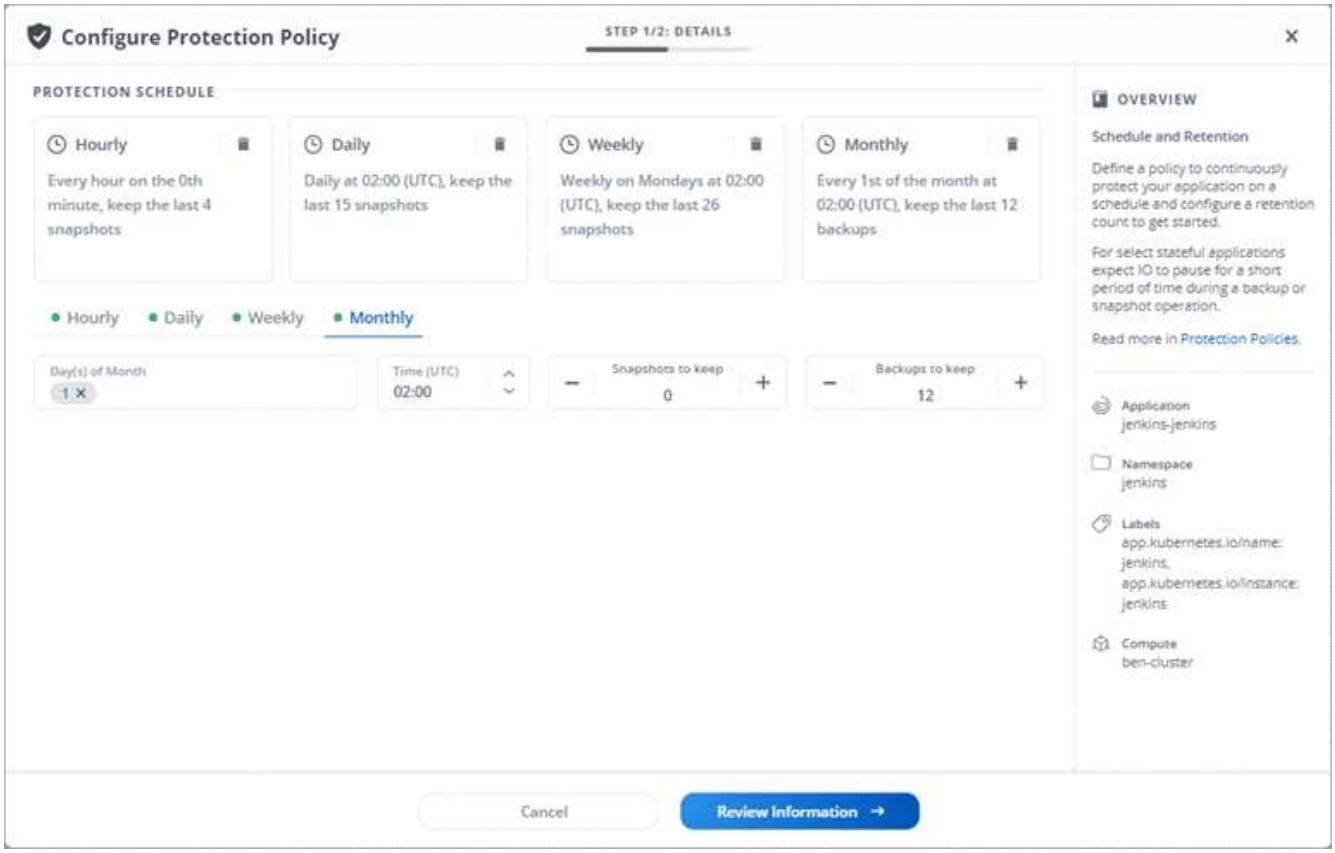
Learn more about Astra Control Service pricing.

- Namespace and app visualization
  We enhanced the Discovered Apps page to better show the hierarchy between namespaces and apps. Just expand a namespace to see the apps contained in that namespace.

  Learn more about managing apps.



- User interface enhancements
  Data protection wizards were enhanced for ease of use. For example, we refined the Protection Policy wizard to more easily view the protection schedule as you define it.



- Activity enhancements
  We've made it easier to view details about the activities in your Astra Control account.
  - Filter the activity list by managed app, severity level, user, and time range.
  - Download your Astra Control account activity to a CSV file.
  - View activities directly from the Clusters page or the Apps page after selecting a cluster or an app.

    Learn more about viewing your account activity.

1 March 2021

Astra Control Service now supports the *CVS* service type with Cloud Volumes Service for Google Cloud. This is in addition to already supporting the *CVS-Performance* service type. Just as a reminder, Astra Control Service uses Cloud Volumes Service for Google Cloud as the storage backend for your persistent volumes.

This enhancement means that Astra Control Service can now manage app data for Kubernetes clusters that are running in *any* Google Cloud region where Cloud Volumes Service is supported.

If you have the flexibility to choose between Google Cloud regions, then you can pick either CVS or CVS-Performance, depending on your performance requirements. Learn more about choosing a service type.

25 January 2021

We're pleased to announce that Astra Control Service is now Generally Available. We incorporated a lot of the feedback that we received from the Beta release and made a few other notable enhancements.

- Billing is now available, which enables you to move from the Free Plan to the Premium Plan. Learn more about billing.

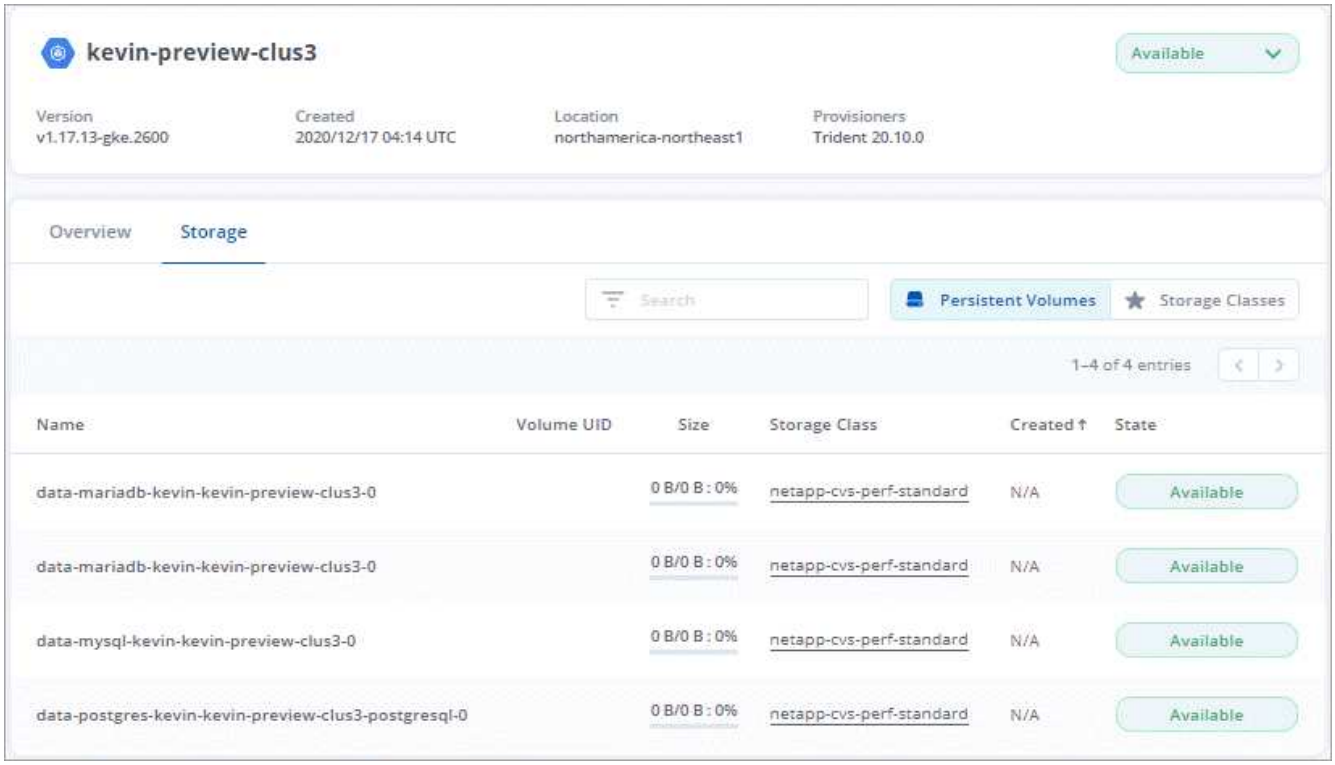
- Astra Control Service now creates Persistent Volumes with a minimum size of 100 GiB when using the CVS-Performance service type.
- Astra Control Service can now discover apps faster.
- You can now create and delete accounts on your own.
- We've improved notifications when Astra Control Service can no longer access a Kubernetes cluster.

  These notifications are important because Astra Control Service can't manage apps for disconnected clusters.

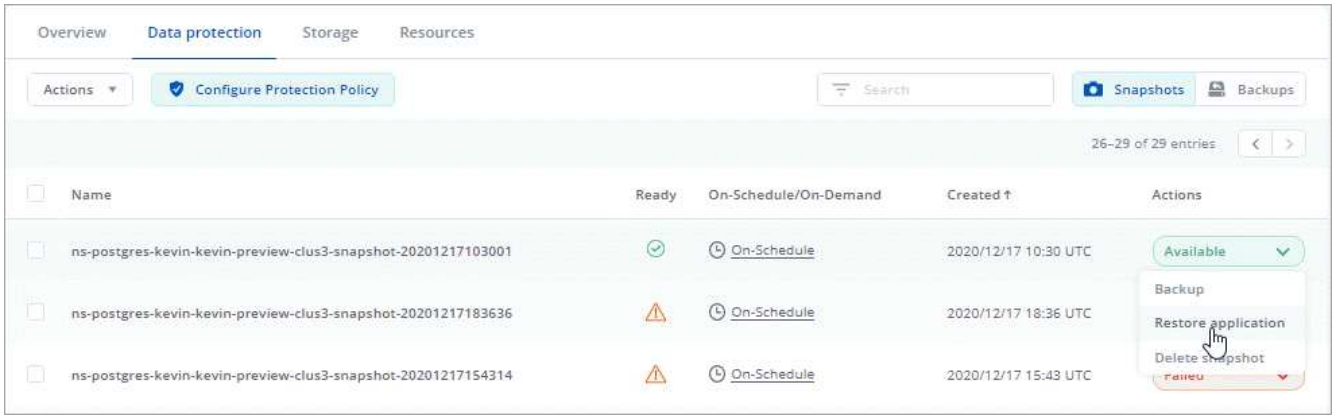
17 December 2020 (Beta update)

We primarily focused on bug fixes to improve your experience, but we made a few other notable enhancements:

- When you add your first Kubernetes compute to Astra Control Service, the object store is now created in the geography where the cluster resides.
- Details about persistent volumes is now available when you view storage details at the compute level.



- We added an option to restore an application from an existing snapshot or backup.



- If you delete a Kubernetes cluster that Astra Control Service is managing, the cluster now shows up in a **Removed** state. You can then remove the cluster from Astra Control Service.
- Account owners can now modify the assigned roles for other users.
- We added a section for billing, which will be enabled when Astra Control Service is released for General Availability (GA).

Known issues

Known issues identify problems that might prevent you from using this release of the product successfully.

The following known issues affect the current release:

Apps

- Unable to define an app on a namespace that has been deleted and recreated

Backup, restore, and clone

- App clones fail using a specific version of PostgreSQL
- App backups and snapshots fail if the volumesnapshotclass is added after a cluster is managed
- In-place restore operations to ontap-nas-economy storage classes fail
- Restoring from a backup when using Kerberos in-flight encryption can fail
- Backup data remains in bucket after deletion for buckets with expired retention policy

Other issues

- App data management operations fail with Internal Service Error (500) when Astra Trident is offline

Unable to define an app on a namespace that has been deleted and recreated

If you define an application with a namespace, delete the namespace, and then reinstall the app in the same namespace, the operation fails with a 409 error code. To define the app using the recreated namespace, delete the old application instance first.

App clones fail using a specific version of PostgreSQL

App clones within the same cluster consistently fail with the Bitnami PostgreSQL 11.5.0 chart. To clone successfully, use an earlier or later version of the chart.

App backups and snapshots fail if the volumesnapshotclass is added after a cluster is managed

Backups and snapshots fail with a UI 500 error in this scenario. As a workaround, refresh the app list.

In-place restore operations to ontap-nas-economy storage classes fail

If you perform an in-place restore of an application (restore the app to its original namespace), and the app's storage class uses the `ontap-nas-economy` driver, the restore operation can fail if the snapshot directory is not hidden. Before restoring in-place, follow the instructions in Enable backup and restore for ontap-nas-economy operations to hide the snapshot directory.

Restoring from a backup when using Kerberos in-flight encryption can fail

When you restore an application from a backup to a storage backend that is using Kerberos in-flight encryption, the restore operation can fail. This issue does not affect restoring from a snapshot or replicating the application data using NetApp SnapMirror.

> When using Kerberos in-flight encryption with NFSv4 volumes, ensure that the NFSv4 volumes are using the correct settings. Refer to the NetApp NFSv4 Domain Configuration section (page 13) of the NetApp NFSv4 Enhancements and Best Practices Guide.

Backup data remains in bucket after deletion for buckets with expired retention policy

If you delete an app's immutable backup after the bucket's retention policy has expired, the backup is deleted from Astra Control but not from the bucket. This issue will be fixed in an upcoming release.

App data management operations fail with Internal Service Error (500) when Astra Trident is offline

If Astra Trident on an app cluster goes offline (and is brought back online) and 500 internal service errors are encountered when attempting app data management, restart all of the Kubernetes nodes in the app cluster to restore functionality.

Known limitations

Known limitations identify platforms, devices, or functions that are not supported by this release of the product, or that do not interoperate correctly with it. Review these limitations carefully.

General limitations

The following limitations affect Astra Control Service's management of Kubernetes clusters in any supported Kubernetes deployment.

Existing connections to a Postgres pod causes failures

When you perform operations on Postgres pods, you shouldn't connect directly within the pod to use the psql command. Astra Control Service requires psql access to freeze and thaw the databases. If there is a pre-existing connection, the snapshot, backup, or clone will fail.

The Activity page displays up to 100,000 events

The Astra Control Activity page can display up to 100,000 events. To view all logged events, retrieve the events using the Astra Control REST API.

Limitations for management of GKE clusters

The following limitations apply to the management of Kubernetes clusters in Google Kubernetes Engine (GKE).

App management limitations

The following limitations affect Astra Control Service's management of applications.

Multiple applications that use the same namespace cannot be restored collectively to a different namespace

If you manage multiple applications that use the same namespace (by creating multiple app definitions in Astra Control), you cannot restore all of the applications to a different single namespace. You need to restore each application to its own separate namespace.

Astra Control does not automatically assign default buckets for cloud instances

Astra Control does not automatically assign a default bucket for any cloud instance. You need to manually set a default bucket for a cloud instance. If a default bucket is not set, you won't be able to perform app clone operations between two clusters.

In-place restore operations of apps that use a certificate manager are not supported

This release of Astra Control Service does not support in-place restore of apps with certificate managers. Restore operations to a different namespace and clone operations are supported.

App clones fail after an application is deployed with a set storage class

After an application is deployed with a storage class explicitly set (for example, `helm install …-set global.storageClass=netapp-cvs-perf-extreme`), subsequent attempts to clone the application require that the target cluster have the originally specified storage class.
Cloning an application with an explicitly set storage class to a cluster that does not have the same storage class will fail. There are no recovery steps in this scenario.

Clones of apps installed using pass by reference operators can fail

Astra Control supports apps installed with namespace-scoped operators. These operators are generally designed with a "pass-by-value" rather than "pass-by-reference" architecture. The following are some operator apps that follow these patterns:

- Apache K8ssandra

> For K8ssandra, in-place restore operations are supported. A restore operation to a new namespace or cluster requires that the original instance of the application to be taken down. This is to ensure that the peer group information carried over does not lead to cross-instance communication. Cloning of the app is not supported.

- Jenkins CI

- Percona XtraDB Cluster

Note that Astra Control might not be able to clone an operator that is designed with a "pass-by-reference" architecture (for example, the CockroachDB operator). During these types of cloning operations, the cloned operator attempts to reference Kubernetes secrets from the source operator despite having its own new secret as part of the cloning process. The clone operation might fail because Astra Control is unaware of the Kubernetes secrets in the source operator.

> During clone operations, apps that need an IngressClass resource or webhooks to function properly must not have those resources already defined on the destination cluster.

Role-based Access Control (RBAC) limitations

The following limitations apply to the way Astra Control limits user access to resources or capabilities.

A user with namespace RBAC constraints can add and unmanage a cluster

A user with namespace RBAC constraints should not be allowed to add or unmanage clusters. Due to a current limitation, Astra does not prevent such users from unmanaging clusters.

A Member user with namespace constraints cannot access cloned or restored apps until an Admin user adds the namespace to the constraint

Any `member` user with RBAC constraints by namespace name/ID can clone or restore an app to a new namespace on the same cluster or to any other cluster in their organization's account. However, the same user cannot access the cloned or restored app in the new namespace. After a clone or restore operation creates a new namespace, the account admin/owner can edit the `member` user account and update role constraints for the affected user to grant access to the new namespace.

Snapshots might fail for Kubernetes 1.25 or later clusters with certain snapshot controller versions

Snapshots for Kubernetes clusters running version 1.25 or later can fail if version v1beta1 of the snapshot controller APIs are installed on the cluster.

As a workaround, do the following when upgrading existing Kubernetes 1.25 or later installations:

1. Remove any existing Snapshot CRDs and any existing snapshot controller.
2. Uninstall Astra Trident.
3. Install the snapshot CRDs and the snapshot controller.
4. Install the latest Astra Trident version.
5. Create a VolumeSnapshotClass.

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